



Bridge to Graduate Mathematics

(Expanded with Historical & Philosophical Context)

8-month integrated continuation of your core Analysis and Algebra plan.

Phase I – Topology and the Shape of Thought (Weeks 1–8)

The birth of continuity and abstraction.

Week	Focus	Core Texts	Historical & Philosophical Readings	Key Question
1	Sets, logic, and the emergence of topology	Munkres Ch. 1–2	Cantor, “Contributions to the Founding of the Theory of Transfinite Numbers”; Dedekind, <i>Continuity and Irrational Numbers</i> ; Ferreirós, <i>Labyrinth of Thought</i> Ch. 3; Dauben, <i>Georg Cantor: His Mathematics and Philosophy of the Infinite</i>	How did the infinite become rigorous?
2	Open and closed sets, bases	Munkres Ch. 3–4	Hausdorff, <i>Grundzüge der Mengenlehre</i> (1914); Weyl, <i>Philosophy of Mathematics and Natural Science</i> Ch. 1; Shapiro, <i>Thinking About Mathematics</i> Ch. 5	What is the structure of “space”?
3	Continuity, product spaces, and mapping	Munkres Ch. 5–8	Poincaré, <i>Science and Hypothesis</i> §§2–4; Eves, <i>Foundations and Fundamental Concepts of Mathematics</i> Ch. 9	How did topology generalize continuity?
4	Compactness and the infinite	Munkres Ch. 27–29	Tychonoff (1930); Kline, <i>Mathematical Thought from Ancient to Modern Times</i> Ch. 44; Eilenberg–Steenrod, <i>Foundations of Algebraic Topology</i> Preface	Why is compactness the modern descendant of finiteness?
5	Connectedness and separation	Munkres Ch. 23–25	Kuratowski, <i>Topologie</i> (1933); Bourbaki, <i>General Topology</i> Vol. I Introduction	How do we formalize “wholeness”?
6	Metric spaces and equivalence	Munkres Ch. 43–45	Freudenthal, “The Concept of Space in Topology” (1940); Lakatos, <i>Proofs and Refutations</i> selections	How does metric geometry become structural?
7	Topological groups & manifolds	Lee Ch. 1–3	Klein, <i>Erlangen Program</i> (1872); Weyl, <i>Space, Time, Matter</i> Ch. 2; Corry, <i>Modern Algebra and the Rise of Mathematical Structures</i> Ch. 5	How did group theory merge with geometry?
8	Synthesis & philosophy of space	—	Kant, <i>Critique of Pure Reason</i> (Aesthetic); Cassirer, <i>Substance and Function</i> Ch. 4; Whitehead, <i>Process and Reality</i> selections	What does topology say about the nature of form?

Phase II – Measure and Integration (Weeks 9–16)

Quantifying the infinite: from length to measure.

Week	Focus	Core Texts	Historical & Philosophical Readings	Key Question
9	Sigma-algebras, measurable sets	Folland Ch. 1 §1–2	Borel, <i>Leçons sur la théorie des fonctions</i> (1898); Lebesgue, <i>Leçons sur l'Intégration</i> (1904); Grattan-Guinness, <i>Development of the Foundations of Mathematical Analysis</i>	How did measure overcome paradoxes of infinity?
10	Lebesgue integration	Folland Ch. 2 §1–3; Royden Ch. 4	Lebesgue, “Sur une généralisation de l'intégrale définie”; Hardy, <i>A Course of Pure Mathematics</i> Ch. 6; Ewald, <i>From Kant to Hilbert</i> Vol. II §§	How does integration redefine “area”?
11	Convergence theorems	Folland Ch. 2 §4–5	Riesz (1909); Vitali (1905); Moore, <i>Zermelo's Axiom of Choice</i> selections	How did rigor refine intuition about limits?
12	L^p spaces and completeness	Folland Ch. 3; Rudin Ch. 6	Banach, <i>Théorie des opérations linéaires</i> (1932); von Neumann, <i>Mathematical Foundations of Quantum Mechanics</i>	How did function spaces revolutionize analysis?
13	Product measures, Fubini & Tonelli	Folland Ch. 2 §6	Tonelli (1909); Fubini (1907); Dieudonné, <i>Foundations of Modern Analysis</i>	How did multidimensional integration redefine calculus?
14	Modes of convergence & probability	Folland Ch. 3 §3–5	Kolmogorov, <i>Foundations of the Theory of Probability</i> (1933); Wiener, <i>Cybernetics</i> selections; Uffink, <i>The Principle of Maximum Entropy</i>	What unites randomness and measure?
15	Outer measure & Hausdorff dimension	Folland Ch. 1 §3	Carathéodory (1914); Mandelbrot, <i>The Fractal Geometry of Nature</i> (1982)	How did measure evolve into geometry?
16	Synthesis	—	Poincaré, “The Value of Science” §§3–5; Courant & Robbins, <i>What Is Mathematics?</i>	What does measure say about reality and rigor?

Phase III – Complex Analysis and the Structure of the Plane (Weeks 17–24)

Holomorphicity, transformation, and unity.

Week	Focus	Core Texts	Historical & Philosophical Readings	Key Question
17	Complex functions & differentiation	Ahlfors Ch. 1–2; Stein–Shakarchi Ch.	Euler <i>Introductio in Analysin Infinitorum</i> (1748); Cauchy (1825); Kline Ch. 35	How does complex differentiation transcend the real?

Week	Focus	Core Texts	Historical & Philosophical Readings	Key Question
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18	Cauchy–Riemann & contour integration	Ahlfors Ch. 2–3	Cauchy <i>Cours d’Analyse</i> (1821); Riemann (1851); Bottazzini, <i>Analytic and Projective Geometry</i>	What is analyticity?
19	Cauchy theorem & integral formula	Ahlfors Ch. 3–4	Weierstrass (1869); Dedekind–Weber <i>Über die Theorie der algebraischen Funktionen</i>	How did function theory achieve rigor?
20	Series & residues	Ahlfors Ch. 5–6	Laurent (1843); Mittag-Leffler (1884); Lakatos <i>Proofs and Refutations</i> §§	How did series reveal the structure of singularities?
21	Meromorphic functions & Riemann surfaces	Ahlfors Ch. 6–7	Riemann (1851); Poincaré (1882); Weyl, <i>Die Idee der Riemannschen Fläche</i> (1913)	How do complex structures globalize?
22	Conformal maps	Ahlfors Ch. 8	Schwarz (1869); Koebe (1907); Bell, <i>The Continuous, the Discrete, and the Infinitesimal</i>	What does it mean to preserve form?
23	Harmonic & potential theory	Ahlfors Ch. 9; Stein–Shakarchi Ch. 5	Dirichlet (1829); Hilbert <i>Integral Equations</i> (1904); Batterman <i>The Devil in the Details</i>	How does analysis model the physical world?
24	Synthesis & philosophical reflection	—	Cassirer, <i>The Problem of Knowledge</i> Ch. 4; Whitehead, <i>Science and the Modern World</i> Ch. 6	What is the meaning of the analytic?

Supplementary Contextual Works

Theme	Readings
Philosophy of mathematics	Cassirer <i>Substance and Function</i> ; Husserl <i>Origin of Geometry</i> ; Shapiro <i>Thinking About Mathematics</i> ; Stein <i>On the Concept of Meaning in Mathematics</i>
Mathematics and modern thought	Kline <i>Mathematical Thought from Ancient to Modern Times</i> ; Corry & Gray <i>The Architecture of Modern Mathematics</i> ; Lakatos <i>Proofs and Refutations</i>
Foundations and infinity	Dauben <i>Georg Cantor</i> ; Ferreirós <i>Labyrinth of Thought</i> ; Ewald <i>From Kant to Hilbert</i> Vol. II
Science and philosophy	Weyl <i>Philosophy of Mathematics and Natural Science</i> ; Poincaré <i>Science and Hypothesis</i> ; Whitehead <i>Process and Reality</i> ; Cassirer <i>The Problem of Knowledge</i>
Bridging to modern logic and computation	Kleene <i>Mathematical Logic</i> ; Turing <i>On Computable Numbers</i> ; Sieg <i>Gödel on Concepts</i> ; Mac Lane <i>Mathematics: Form and Function</i>



Integration Tip

To keep balance with the heavy mathematical content:

- Dedicate **Fridays** or **Sundays** to reading and reflecting on the contextual works.
- Aim to **connect** each week's core concept (compactness, convergence, holomorphicity) with its historical and philosophical emergence.
- Keep a *mathematical journal* — each week write 1 page: “How did this concept emerge historically, and what philosophical problem did it solve?”



Bibliography — Bridge to Graduate Mathematics

I. Core Mathematical Texts

Topology

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3. **Willard, Stephen.** *General Topology*. Dover, 2004.
4. **Kelley, John L.** *General Topology*. Springer, 1975.
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6. **Bourbaki, Nicolas.** *General Topology*. Springer, 1989.

Measure and Integration

7. **Folland, Gerald B.** *Real Analysis: Modern Techniques and Their Applications*. 2nd ed. Wiley, 1999.
8. **Royden, H. L., and P. M. Fitzpatrick.** *Real Analysis*. 4th ed. Pearson, 2010.
9. **Rudin, Walter.** *Real and Complex Analysis*. 3rd ed. McGraw–Hill, 1987.
10. **Conway, John B.** *A Course in Functional Analysis*. 2nd ed. Springer, 1990.
11. **Kolmogorov, A. N., and S. V. Fomin.** *Introductory Real Analysis*. Dover, 1975.
12. **Royden, H. L.** *Real Analysis*. 3rd ed. Macmillan, 1988.
13. **Mattila, Pertti.** *Geometry of Sets and Measures in Euclidean Spaces*. Cambridge University Press, 1995.
14. **Durrett, Rick.** *Probability: Theory and Examples*. 5th ed. Cambridge University Press, 2019.

Complex Analysis

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16. **Stein, Elias M., and Rami Shakarchi.** *Complex Analysis*. Princeton University Press, 2003.

17. **Conway, John B.** *Functions of One Complex Variable*. 2nd ed. Springer, 1978.
 18. **Palka, Bruce.** *An Introduction to Complex Function Theory*. Springer, 1991.
 19. **Rudin, Walter.** *Principles of Mathematical Analysis*. 3rd ed. McGraw–Hill, 1976.
 20. **Courant, Richard, and Herbert Robbins.** *What Is Mathematics?* Oxford University Press, 1996.
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II. Primary Historical Sources

Topology

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2. **Richard Dedekind.** *Continuity and Irrational Numbers*. 1872.
3. **Felix Hausdorff.** *Grundzüge der Mengenlehre*. Leipzig, 1914.
4. **Henri Poincaré.** *Science and Hypothesis*. Paris, 1902.
5. **Felix Klein.** *Erlanger Programm*. Erlangen, 1872.
6. **Hermann Weyl.** *Raum, Zeit, Materie*. Berlin, 1918.

Measure Theory

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8. **Henri Lebesgue.** *Leçons sur l'Intégration et la Recherche des Fonctions Primitives*. Paris, 1904.
9. **Giuseppe Vitali.** “Sopra alcune proprietà dei insiemi di punti.” *Rendiconti del Circolo Matematico di Palermo* 17 (1903).
10. **Frigyes Riesz.** “Sur les opérations fonctionnelles linéaires.” *Comptes Rendus de l'Académie des Sciences*, 1909.
11. **Stefan Banach.** *Théorie des opérations linéaires*. Warsaw, 1932.
12. **Constantin Carathéodory.** “Über das lineare Mass von Punktmengen.” *Mathematische Annalen* 67 (1909).
13. **A. N. Kolmogorov.** *Foundations of the Theory of Probability*. Moscow, 1933.
14. **Norbert Wiener.** *The Fourier Integral and Certain of Its Applications*. Cambridge, 1933.

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16. **Augustin-Louis Cauchy.** *Cours d'Analyse*. Paris, 1821.
17. **Bernhard Riemann.** *Grundlagen für eine allgemeine Theorie der Functionen einer veränderlichen complexen Größe*. Göttingen, 1851.
18. **Karl Weierstrass.** *Collected Works*. Vol. 2, 1869–71.

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 21. **Hermann Weyl.** *Die Idee der Riemannschen Fläche.* Teubner, 1913.
 22. **Henri Poincaré.** "Analysis Situs." *Journal de l'École Polytechnique* (1895).
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III. Historical & Interpretive Studies

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3. **Ferreirós, José.** *Labyrinth of Thought: A History of Set Theory and Its Role in Modern Mathematics.* 2nd ed. Birkhäuser, 2007.
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 4. **Alfred North Whitehead.** *Process and Reality*. Macmillan, 1929.
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 11. **Corfield, David.** *Towards a Philosophy of Real Mathematics*. Cambridge University Press, 2003.
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V. Supplemental & Modern Connections

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2. **Lawvere, F. William.** "Functorial Semantics of Algebraic Theories." *Proceedings of the National Academy of Sciences* 50 (1963): 869–872.
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